DataOps Innovation Hackathon 2026 · Problem Statement 2

DocForensics

AI-powered document forensics and contextual intelligence



V-002 · salary slip, net pay

Potential alteration · confidence high (0.90)

Located on the page. Never labelled as fraud.

Arnav Modi · MSIT

Reading a document is not trusting it

SALARY SLIP

Net pay

Rs. 85,000

OCR reads it correctly

≠

No single-document check sees this

BANK STATEMENT

Salary credit

Rs. 58,000

OCR reads it correctly

DocForensics adds the missing step: is a value altered, and do the documents agree?

Architecture: four stages, one audit trail

Input: scans · phone photos · PDFs · ZIP bundles

1 PROCESS

SHA-256 · deskew

Text layer or OCR

Extract fields

Classify · 5 types

Tesseract · PyMuPDF · bge-small

2 LOOK

Scans: ELA, noise, tone, ink, copy-move

PDFs: fonts, patches, hidden text, file history

OpenCV · NumPy

3 REASON

Arithmetic · dates · IDs

Cross-document matching

Income reconciliation

Rule engine

4 EVIDENCE

Boxed evidence crops

Dossier PDF + JSON

Reviewer decisions

Local files

Hash-chained audit log: every run, finding and reviewer decision

Visual forensics: one question, two branches

What kind of file?

Scan or photo

ELA · Noise · Tone · Ink darkness · Typography · Copy-move

Each value is compared with its peers on the same page



Scan: net pay flagged

Generated PDF

Font outliers · Patched blocks · Hidden text · Revisions · Info vs XMP · Font subsets

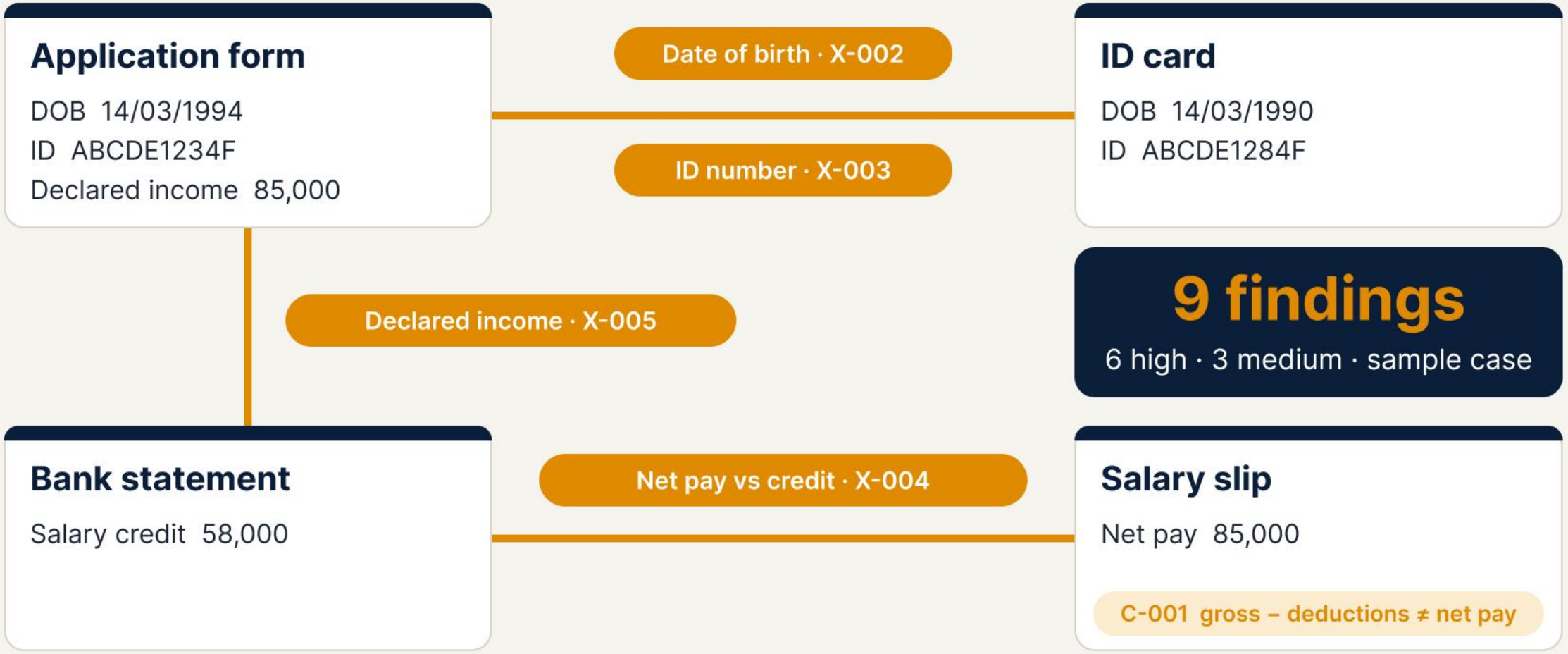
File-history signals are low-confidence observations



PDF: new typeface, patched block

Checks that read the file or compare values, not a score from a black box.

Cross-document reasoning: the files audit each other



Arithmetic, identity and income checks do not depend on how an edit was made.

Every finding traces back to the file

Finding — ID · status · score → **Fields** — values it rests on → **Box** — [x0, y0, x1, y1] → **Page crop** — zoomed evidence → **File** — SHA-256 of original → **Audit entry** — hash-chained log

7.5 Evidence and Traceability

Each finding is tied to its original documents (file hash), the extracted fields and values it rests on, and the region it points to (box as in 7.2). Zoomed crops are printed with each finding in 7.3 and 7.4; the highlighted pages follow the table.

ID	Documents [SHA-256 first 8]	Associated extracted information	Region [x0,y0,x1,y1]	Status	Confidence
C-001	salary_slip[7ef5677d]	salary_slip.gross=Rs. 95,000; salary_slip.deductions=Rs. 37,000; salary_slip.net_pay=Rs. 85,000	[433, 451, 578, 475]; [433, 536, 578, 560]; [432, 625, 579, 648]	inconsistency	high (0.98)
X-002	application_form[b65f9da5], id_card[c47d6e90]	id_card.dob=14/03/1990; application_form.dob=14/03/1994	[434, 285, 590, 308]; [433, 281, 590, 304]	inconsistency	high (0.95)
X-004	bank_statement[4805d814], salary_slip[7ef5677d]	salary_slip.net_pay=Rs. 85,000; bank_statement.salary_credit=Rs. 58,000	[432, 625, 579, 648]; [433, 536, 578, 560]	inconsistency	high (0.95)
V-001	id_card[c47d6e90]	id_card.dob=14/03/1990	[434, 285, 590, 308]	potential manipulation	high (0.90)
V-002	salary_slip[7ef5677d]	salary_slip.net_pay=Rs. 85,000	[432, 625, 579, 648]	potential manipulation	high (0.90)
X-003	application_form[b65f9da5], id_card[c47d6e90]	id_card.id_number=ABCDE1284F; application_form.id_number=ABCDE1234F	[430, 366, 613, 386]; [430, 366, 613, 386]	inconsistency	high (0.90)
X-001	application_form[b65f9da5], bank_statement[4805d814], id_card[c47d6e90], salary_slip[7ef5677d]	bank_statement.name=Rohan Mehra; application_form.name=Rohan Mehta; id_card.name=Rohan Mehta; salary_slip.name=Rohan Mehta	[433, 195, 613, 216]; [433, 195, 612, 216]; [433, 195, 612, 216]; [433, 195, 612, 216]	inconsistency	high (0.80)
X-005	application_form[b65f9da5], bank_statement[4805d814]	application_form.declared_income=85,000; bank_statement.salary_credit=Rs. 58,000	[484, 621, 578, 645]; [433, 536, 578, 560]	inconsistency	medium (0.70)
R-001	bank_statement[4805d814]	(whole page)	-	observed	medium (0.60)

Status

- **observed**
- **inconsistency**
- **potential manipulation**

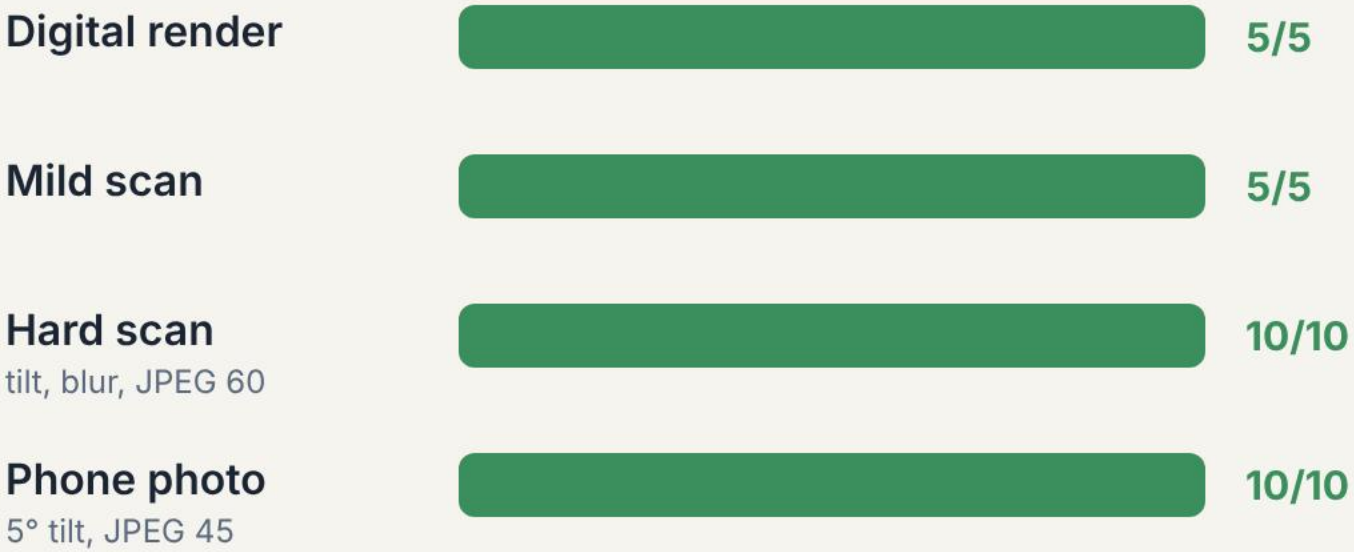
Confidence

high ≥ 0.80 · medium 0.60–0.79 · low < 0.60

The dossier is system-generated. Reviewer decisions are stored separately in the audit log.

Results, stated honestly

Naive forgeries detected (pasted value, different ink or tone)



108/108
fields located

5/5
planted findings

0
false findings, clean controls

170+
automated tests

Not detected: skilled forgeries (font, ink and tone matched, every number consistent): 0 of 8. Nothing in the files contradicts such an edit; it needs issuer verification.

All data is synthetic and self-generated, and thresholds were tuned on it. Expect lower numbers on real documents.

What is built, and what is not

Implemented	Heuristic	Planned	Unsupported
OCR · extraction · 5 types	Typography and layout	Signature authenticity	Handwriting
Scan and PDF forensics	Signature placement	JPEG-ghost check	
Cross-document reasoning		Hindi / multilingual OCR	
Dossier · audit · UI		Layout-aware models	

Known limits

Skilled forgeries evade pixel checks (0 of 8 in the benchmark)

A pasted copy recompressed below JPEG 92 can be missed

Findings are indicators for a reviewer, not proof

Demo: one finding, end to end

1. Upload a synthetic case
2. Read the findings table
3. Open one finding: crop, box, page
4. Trace it to SHA-256 and audit entry
5. Download dossier, record a decision

Repository

github.com/ArnavModi-MSIT/DocForensics-Investigator

Code · README · architecture notes · sample dossiers

Synthetic documents only. No real personal data is used or shown.

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